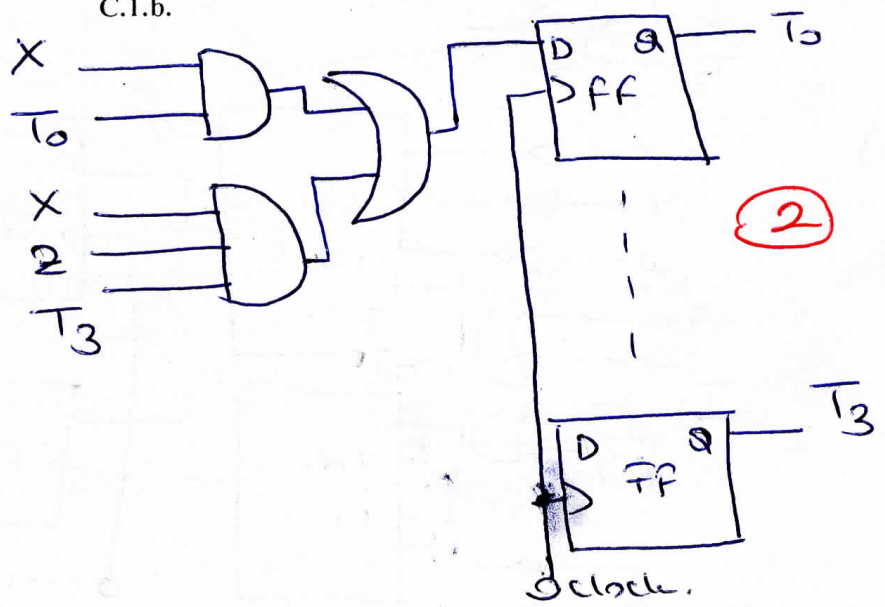




S.2. Birinci soruda istenen kontrol ünitesini

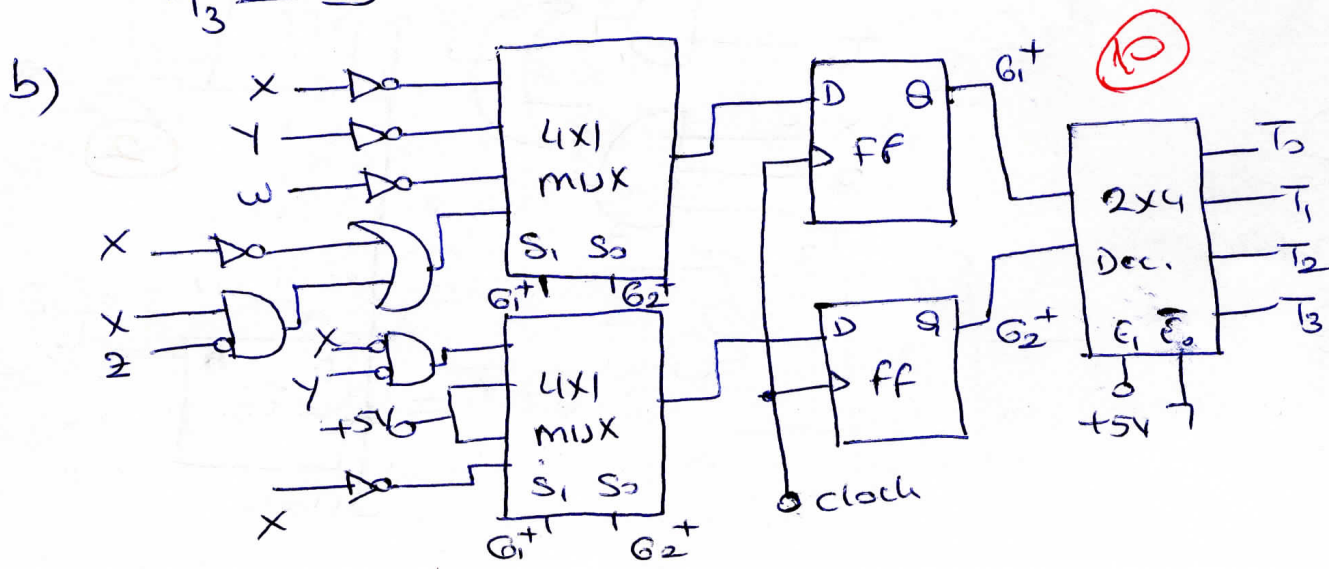
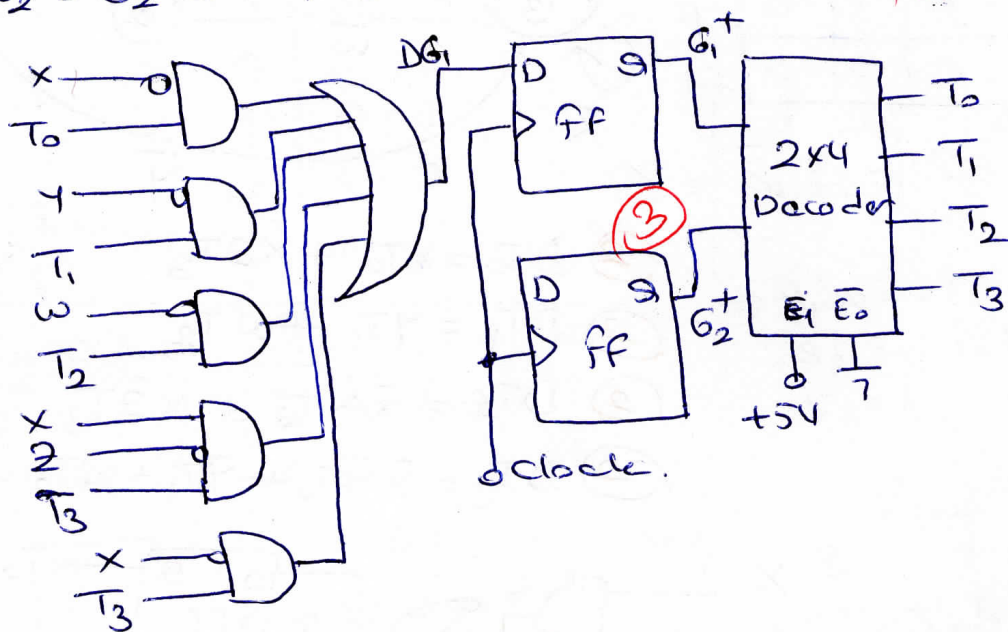
a) FF (D-FF) ve Kod Çözücü (Decoder) kullanarak tasarlayınız. (7p+8p(Tablo))

b) MUX, FF (D-FF) ve Kod Çözücü (Decoder) kullanılarak tasarlayınız. (10p)

S.D	S.D	G.D			
Sembol	$G_1 G_2$	$G_1 G_2$	Giriz	MUX-1	MUX-2
T_0	0 0	0 0	X	$= \bar{X}Y + \bar{X}\bar{Y}$	$\bar{X}\bar{Y}$
T_0	0 0	1 0	$\bar{X}Y$	$= \bar{X}$	
T_0	0 0	1 1	$\bar{X}\bar{Y}$		
T_1	0 1	0 1	Y	$\bar{Y}$	1
T_1	0 1	1 1	$\bar{Y}$		
T_2	1 0	0 1	W	$\bar{W}$	1
T_2	1 0	1 1	$\bar{W}$		
T_3	1 1	0 0	$X\bar{Z}$	$\bar{X} + X\bar{Z}$	$\bar{X}$
T_3	1 1	1 0	$X\bar{Z}$		
T_3	1 1	1 1	$\bar{X}$		

a) $DG_1 = G_1^+ = \bar{X}T_0 + \bar{Y}T_1 + \bar{W}T_2 + (X\bar{Z} + \bar{X})T_3$ (2)

$DG_2 = G_2^+ = \bar{X}\bar{Y}T_0 + T_1 + T_2 + \bar{X}T_3$ (2)



S.3. Birinci soruda istenen kontrol ünitesini PLA kullanarak tasarlayınız. PLA genel şemasını ve T_0 için PLA açık devresini çiziniz.

(10p) NOT: PLA' da gerçekleştirilmesi gereken bütün çıkışlara ait lojik fonksiyonları yazınız.

C.3.

$$\bar{G}_1 = \bar{X}\bar{Y}\bar{G}_2 + \bar{Y}\bar{G}_1\bar{G}_2 + \bar{W}\bar{G}_1\bar{G}_2 + (\bar{X}Z + \bar{X})\bar{G}_1\bar{G}_2$$

$$G_2^+ = \bar{X}\bar{Y}\bar{G}_1\bar{G}_2 + \bar{G}_1\bar{G}_2 + G_1\bar{G}_2 + \bar{X}G_1\bar{G}_2 \quad (1)$$

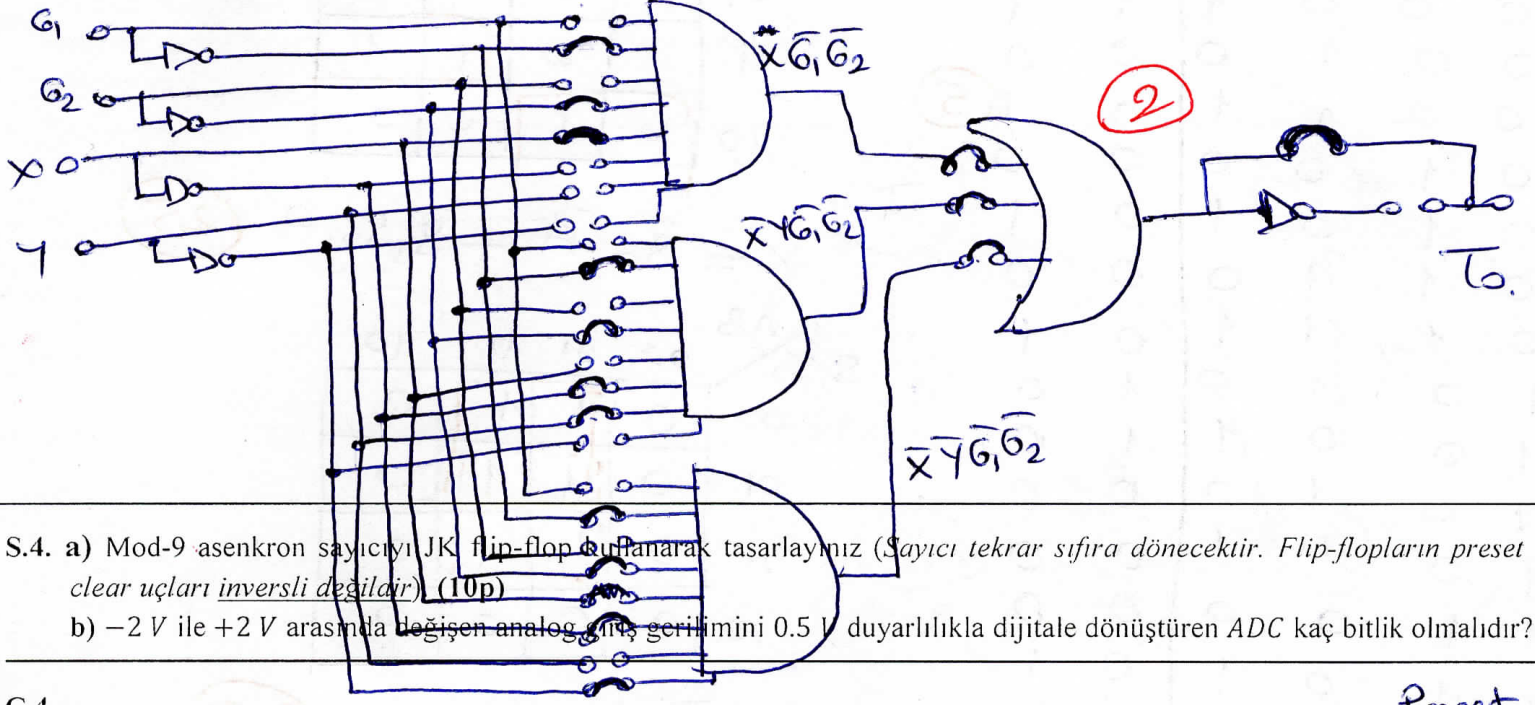
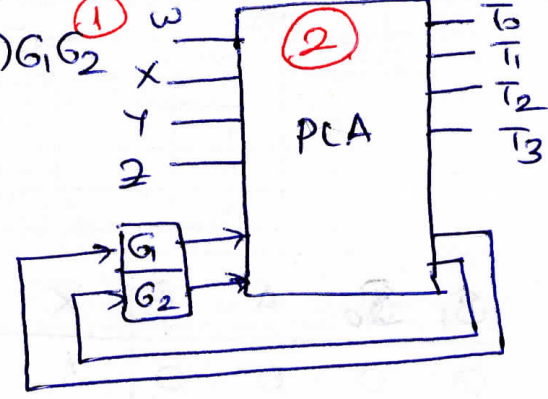
$$G_1^+ = \bar{X}\bar{G}_1\bar{G}_2 + \bar{Y}\bar{G}_1\bar{G}_2 + \bar{W}\bar{G}_1\bar{G}_2 + (\bar{X}Z + \bar{X})\bar{G}_1\bar{G}_2 \quad (1)$$

$$T_0 = X\bar{G}_1\bar{G}_2 + \bar{X}Y\bar{G}_1\bar{G}_2 + \bar{X}\bar{Y}\bar{G}_1\bar{G}_2 \quad (1)$$

$$T_1 = Y\bar{G}_1\bar{G}_2 + Y\bar{G}_1\bar{G}_2 \quad (1)$$

$$T_2 = W\bar{G}_1\bar{G}_2 + \bar{W}\bar{G}_1\bar{G}_2 \quad (1)$$

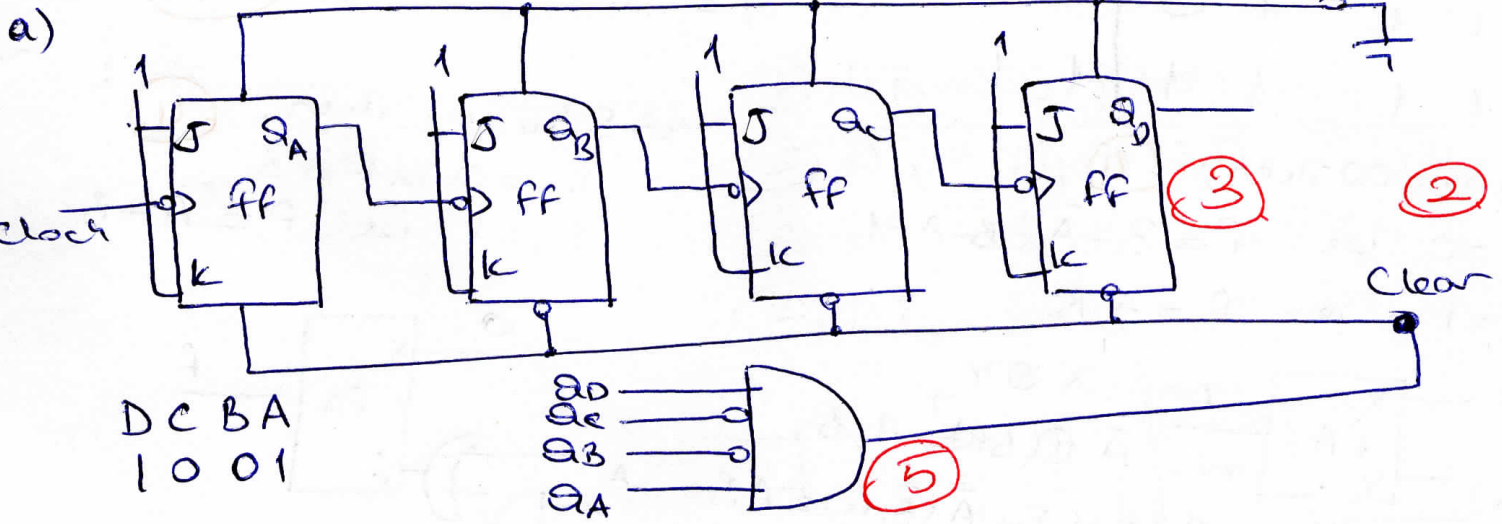
$$\bar{T}_3 = XZ\bar{G}_1\bar{G}_2 + XZ\bar{G}_1\bar{G}_2 + \bar{X}G_1\bar{G}_2 \quad (1)$$



S.4. a) Mod-9 asenkron sayıcıyı JK flip-flop kullanarak tasarlayınız (Sayıcı tekrar sıfıra dönecektir. Flip-flopların preset (set) ve clear uçları inversli değildir). (10p)

b) $-2V$ ile $+2V$ arasında değişen analog giriş gerilimini $0.5V$ duyarlılıkla dijitale dönüştüren ADC kaç bitlik olmalıdır? (10p)

C.4.



b) $a = \text{Duyarlılık} = \frac{V_{\max} - V_{\min}}{2^n}$ $\rightarrow$ ada $(2^n \leq 4)$

$$0,5 = \frac{2 - (-2)}{2^n} \Rightarrow 2^n = 8 \quad \boxed{n = 3 \text{ bit}} \quad (10)$$

S.5. Tabloda verilen işlemleri gerçekleştiren ALU'yu tasarlayınız (Devre çizilmeyecektir). (Aritmetik İşl.: 15p, Lojik İşl.: 15p)

Seçme Girişleri		Aritmetik işlemler ($S_2 = 0$)		Lojik işlemler ($S_2 = 1$)
S_1	S_0	$C_{in} = 0$	$C_{in} = 1$	$C_{in} = 0$
0	0	$F = B - A - 1$	$F = B - A$	$F = A \cdot B$
0	1	$F = B$	$F = B + 1$	$F = A + B$
1	0	$F = \bar{A}$	$F = \bar{A} + 1$	$F = \bar{A} + B$
1	1	$F = A + B$	$F = A + B + 1$	$F = A \oplus B$

X	Y
$\bar{A}$	B
0	B
A	0
A	B

C.5.

S_1	S_0	A	B	X	Y
0	0	0	0	1	0
0	0	0	1	1	1
0	0	1	0	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	0	1	0	1
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	1	0
1	0	0	1	1	0
1	0	1	0	0	0
1	0	1	1	0	0
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	1	0
1	1	1	1	1	1

$S_1 S_0$	00	01	11	10
00	1	1	0	0
01	0	0	0	0
11	0	0	1	1
10	1	1	0	0

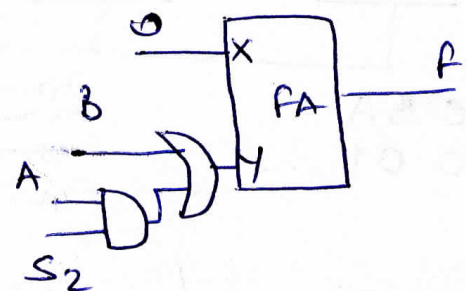
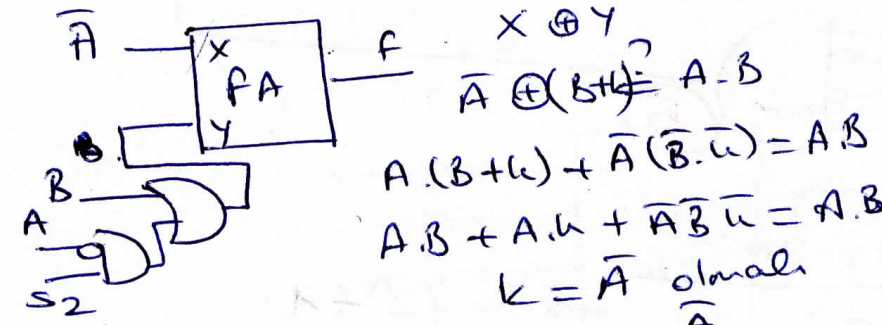
$X = \bar{S}_0 \cdot \bar{A} + S_1 S_0 \cdot A$ (2,5)

$S_1 S_0$	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	1	1	0
10	0	0	0	0

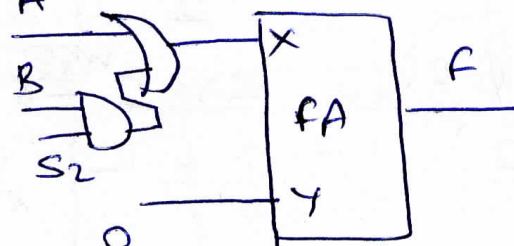
$Y = \bar{S}_1 \cdot B + S_0 \cdot B$ (2,5)

$\Rightarrow S_1 S_0 = 01$ iken; (4)
 $S_2 = 0$ ise $F = B$
 $S_2 = 1$ ise $F = A + B$

$\Rightarrow S_1 S_0 = 00$ iken; (4)
 $S_2 = 0$ ise $F = B + \bar{A} = B - A - 1$
 $S_2 = 1$ ise $F = A \cdot B$



$\Rightarrow S_1 S_0 = 10$ iken (4)
 $S_2 = 0$ ise $F = \bar{A}$
 $S_2 = 1$ ise $F = \bar{A} + B$



$\Rightarrow S_1 S_0 = 11$ iken (3)
 Dönerkenye gerek yok